

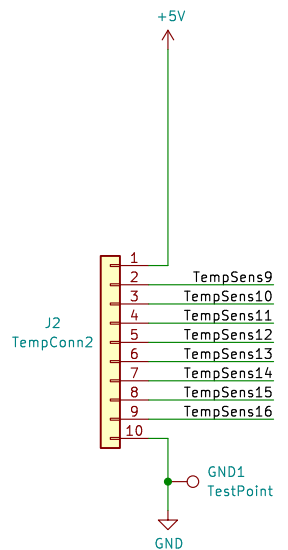
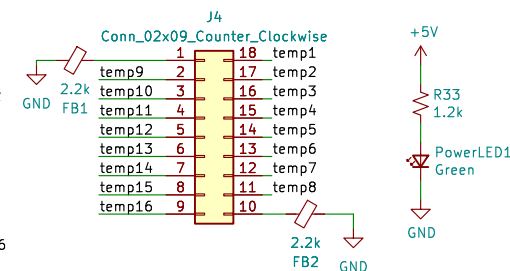
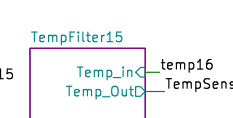
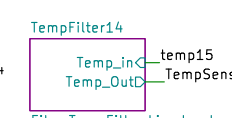
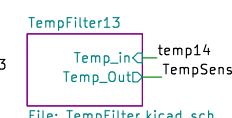
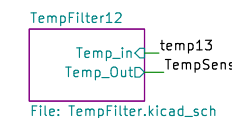
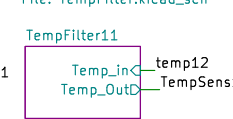
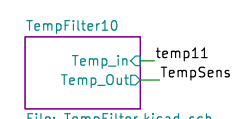
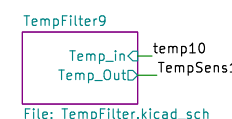
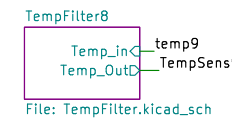
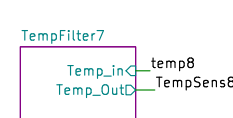
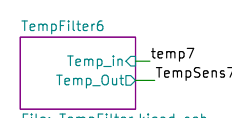
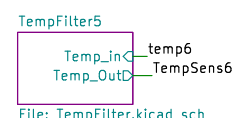
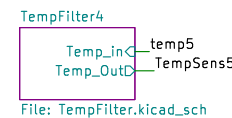
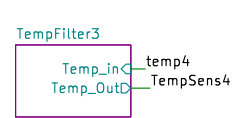
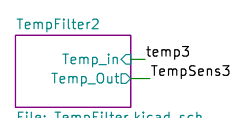
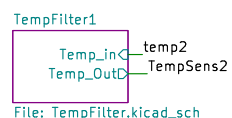
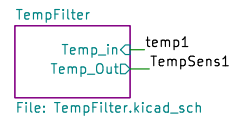
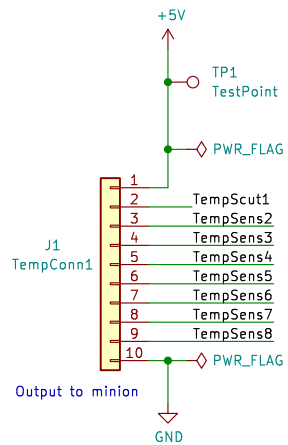


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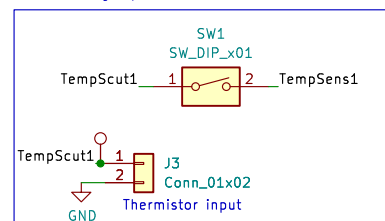
v2.3:
Fix footprint issues
Update RC values to match timing of the LTC6811
Redo PCB layout to reduce islands and vias

v2.2:
PCB layout to make everything symmetrical
and easier to solder

v2.1:
Board supports 16 temperature sensors, add
RC filter to all temperature sensors



Scrutineering Inputs



LTC6811 delay between sleep and measuring ADC voltage is T_WAKE+T_REFUP
 $T_WAKE+T_REFUP = 200\mu s + 2.7ms = 2.9ms$ (minimum)
 $f_{req} = 1 / 0.0029 = 345Hz$

Setting LC cutoff freq to $2 * 345 = 690$
Using a 47nF cap and 4.7k resistor,
we achieve a cutoff freq of 720Hz.

Sheet: /
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Title: MinionShield

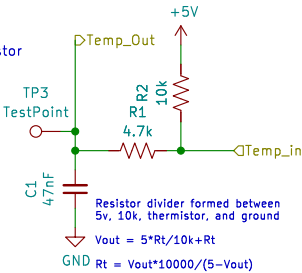
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KiCad E.D.A. 9.0.0

Date: 10/8/22

Rev: 2.2

Id: 1/17

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10,000 ohms



Sheet: /TempFilter/
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Title:

Size: A4

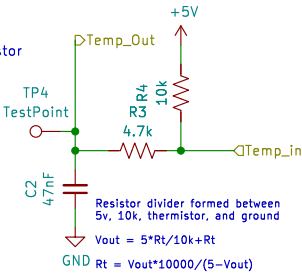
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KiCad E.D.A. 9.0.0

Rev:

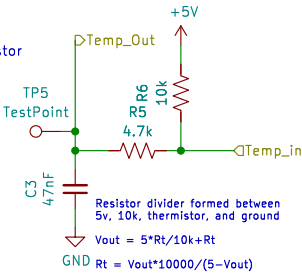
Id: 2/17

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10,000 ohms



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Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.0	Id: 3/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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Size: A4

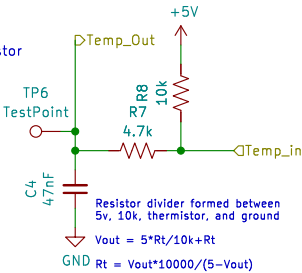
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KiCad E.D.A. 9.0.0

Rev:

Id: 4/17

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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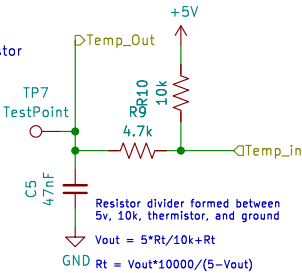
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Rev:

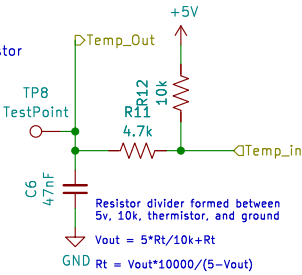
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Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



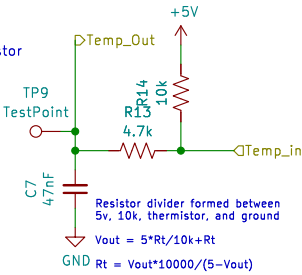
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Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.0	Id: 6/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10,000 ohms



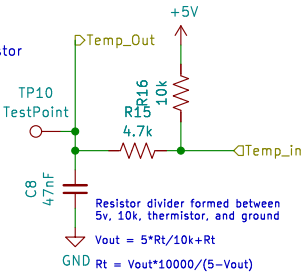
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KiCad E.D.A. 9.0.0	Id: 7/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



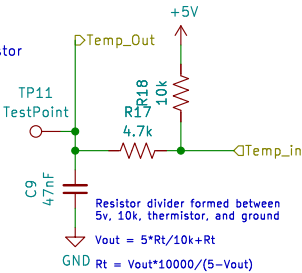
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KiCad E.D.A. 9.0.0	Id: 8/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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KiCad E.D.A. 9.0.0	Id: 9/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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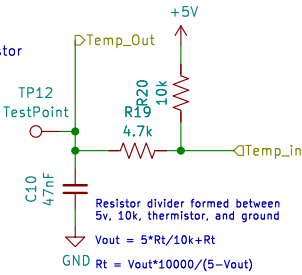
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Rev:

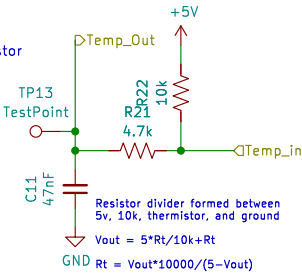
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Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10,000 ohms



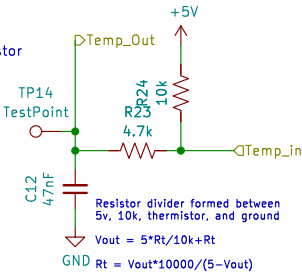
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KiCad E.D.A. 9.0.0	Id: 11/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.0	Id: 12/17	

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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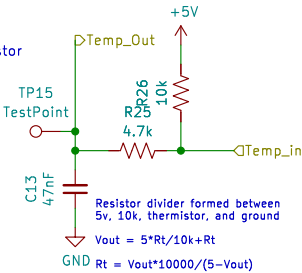
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Rev:

Id: 13/17

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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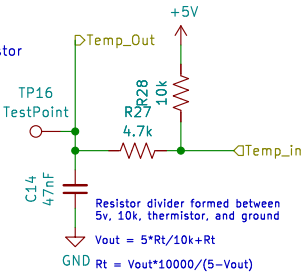
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Rev:

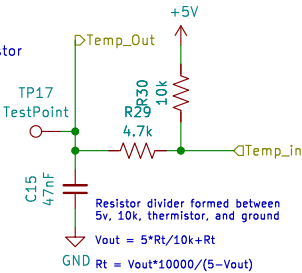
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Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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Size: A4	Date:	Rev:
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Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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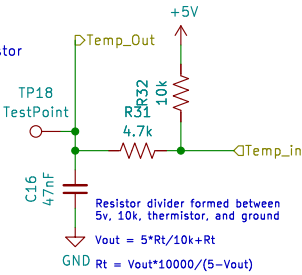
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Rev:

Id: 16/17

Using the CWF1B103F3380 thermistor
With B=3380, R0 = 10.000 ohms



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